



REAL ESTATE FINANCE AND INVESTMENTS

RISKS AND OPPORTUNITIES

Chapter 11

Development Feasibility Analysis

Feasibility Assessment

- Back of the envelope (BOE) analysis should precede detailed pro forma
- Development expected cost summary
- Return on Total Development Cost (“build to” calculation)
 - Expected Stabilized NOI / Expected Total Costs

Feasibility Assessment

- Effective Rent Per Leasable Square Foot Calculation
 - $(\text{Market Rent per Leasable SF} * (1 - \text{Stabilized Vacancy \%}))$
- Stabilized NOI
 - Rental Revenues – Operating Costs

Figure 11.1A

Expected Development Cost and Operating Assumptions for Muk Office Plaza	
Land Cost	\$30 per GSF (building gross square foot)
Hard Costs	\$90 per GSF
<u>Soft Costs</u>	<u>\$30 per GSF</u>
Total Development Cost	\$150 per GSF
Building Size	100,000 GSF
Loss Factor	30% of GSF
Leasable Square Feet	70,000 SF
Stabilized Vacancy	5% of Leasable Square Footage
Occupancy ($1 - \text{Vacancy}$)	95% of Leasable Square Footage
Leased Square Feet	$95\% * 70,000 = 66,500$ SF
Rent	\$30 per Leasable Square Foot
Operating Cost	\$10 per GSF

Figure 11.1B

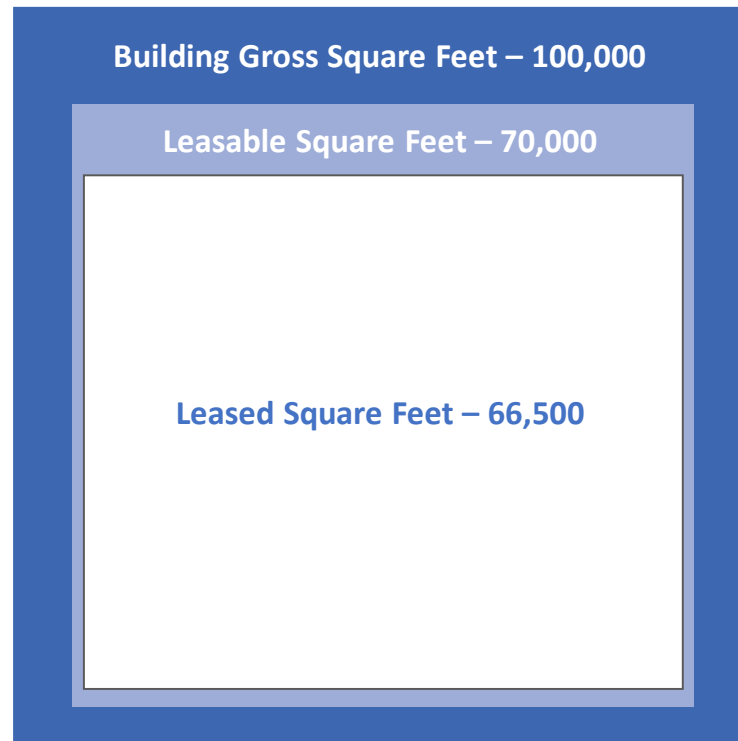


Figure 11.2

Build to x% Return* Calculation

Build to x% Return = Projected Stabilized NOI / Projected Total Costs

Build to a 10% = \$15.00 per GSF / \$150.00 per GSF

** Alternately referred to as the Yield on Cost, Return on Cost, or the Going in Cap Rate*

Figure 11.3

Effective Rent per Leasable Square Foot Calculation

$$\begin{aligned}\text{Effective Rent per Leasable SF} &= \\ \text{Market Rent per Leasable SF} * (1 - \text{Stabilized Vacancy \%}) \\ &= \$30.00 * (1 - .05) \\ &= \$28.50\end{aligned}$$

Figure 11.4

Converting Effective Rent per Leasable Square Foot to Effective Rent per Gross Square Foot

Effective Rent per Gross SF = Effective Rent per Leasable SF * (1 – Loss Factor %)

= \$28.50 * (1 – 0.30)

= \$19.95 per GSF

Figure 11.5

Stabilized NOI per Gross Square Foot Calculation

$$\begin{aligned}\text{Stabilized NOI per GSF} &= \text{Rental Revenues per GSF} - \text{Operating Costs per GSF} \\ &= \$19.95 \text{ PSF} - \$10.00 \text{ PSF} = \$9.95 \text{ PSF}\end{aligned}$$

Figure 11.6

Expected Yield on Total Development Cost Calculation

Expected Yield on Cost = Expected Stabilized NOI / Expected Total Costs

= \$9.95 / \$150.00 = 6.63%, which is less than the required 10.00%

Solve Backwards for Rent

- Replacement Rent per GSF (necessary market rent for feasibility)
 - $(\text{Build to Return} * \text{Expected Total Cost}) + \text{Expected Operating Costs}$
- Replacement Rent per Leasable SF
 - $\text{Replacement Rent per GSF} * (1/(1 - \text{Loss Factor})) * (1/(1 - \text{Vacancy}))$
 - Removes unleasable space and space that will be vacant to back into minimum gross rent to achieve required return

Figure 11.7

Replacement Rent per Gross Square Foot Calculation

$$\begin{aligned}\text{Replacement Rent per GSF} &= (\text{Build to Return} * \text{Expected Total Cost}) + \text{Expected Operating Costs} \\ &= (10\% * \$150.00) + \$10.00 = \$15.00 + \$10.00 = \$25.00 \text{ per GSF}\end{aligned}$$

Figure 11.8

Replacement Rent per Leasable Square Foot Calculation

$$\begin{aligned} &\text{Replacement Rent per Leasable Square Foot} = \\ &\text{Replacement Rent per GSF} * (1 / (1 - \text{Loss Factor})) * (1 / (1 - \text{Vacancy})) \\ &= \$25.00 * (1 / (1 - 0.3)) * (1 / (1 - 0.05)) = \$37.59 \text{ per Leasable Square Foot} \end{aligned}$$

Common Mistakes

- Rental projections
- Cost estimation
- Timing
- Never lose sight of supply & demand fundamentals

Figure 11.9

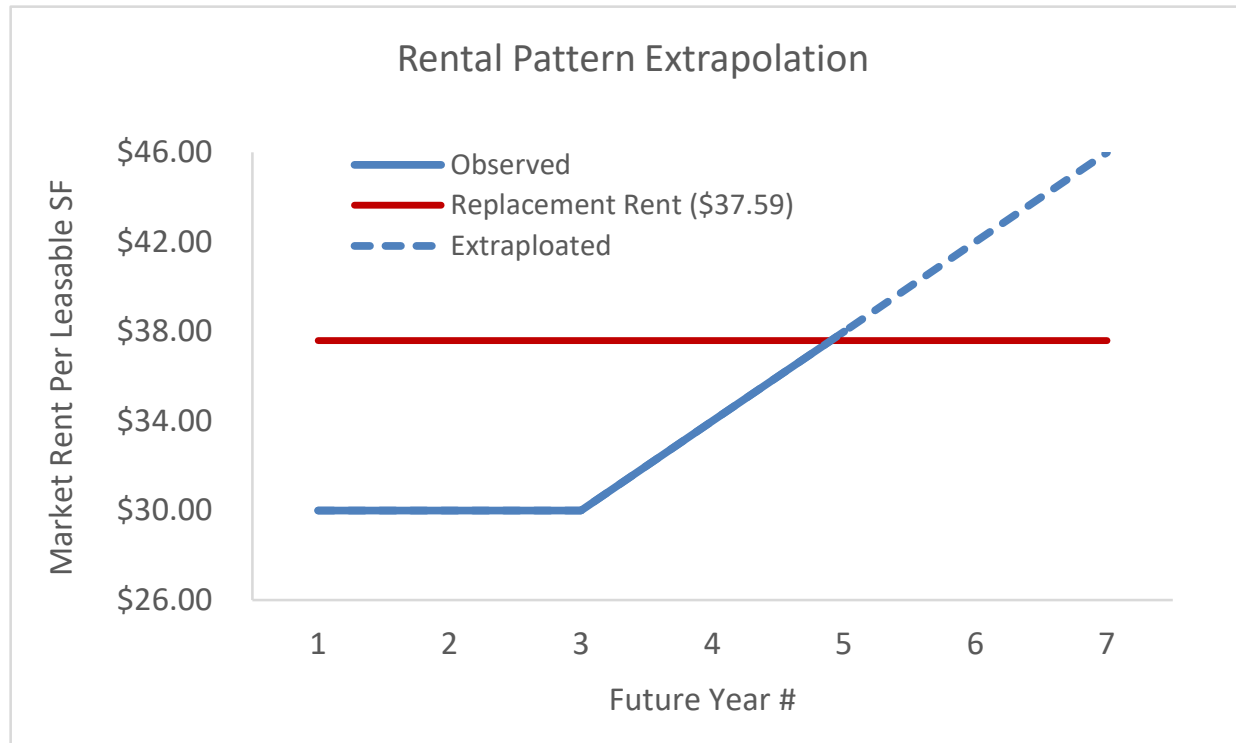


Figure 11.10

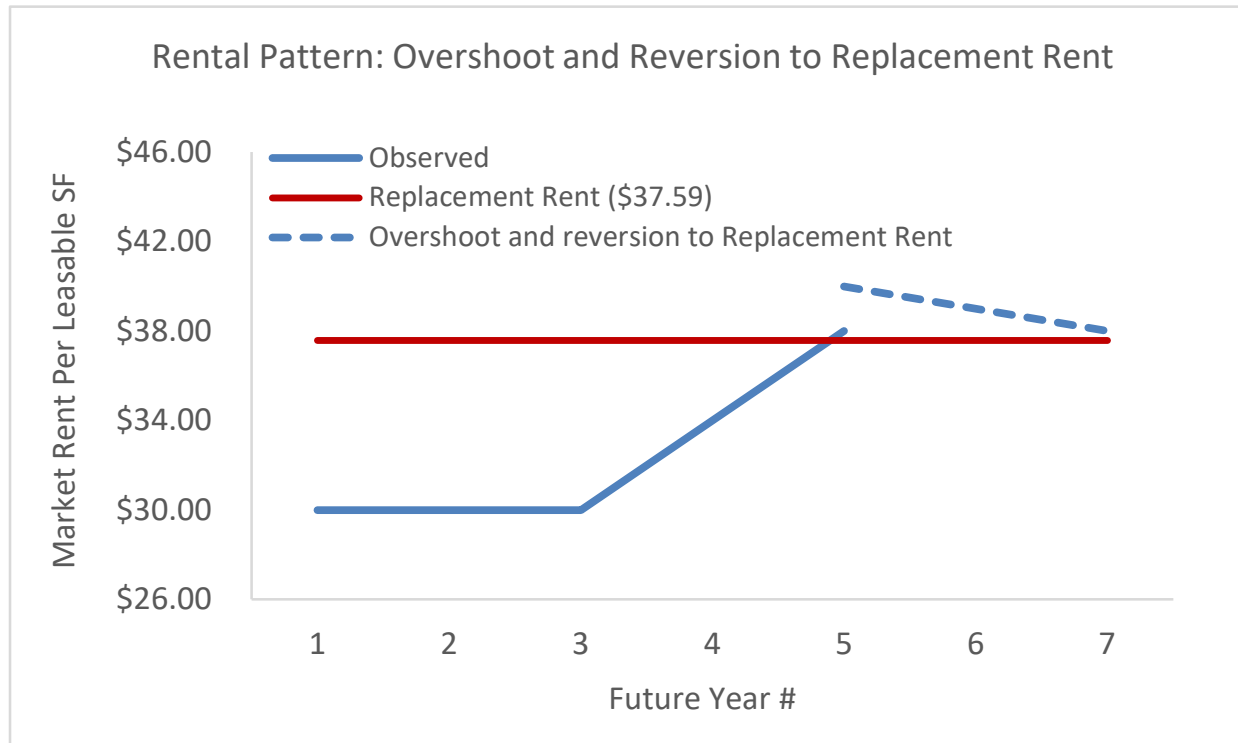
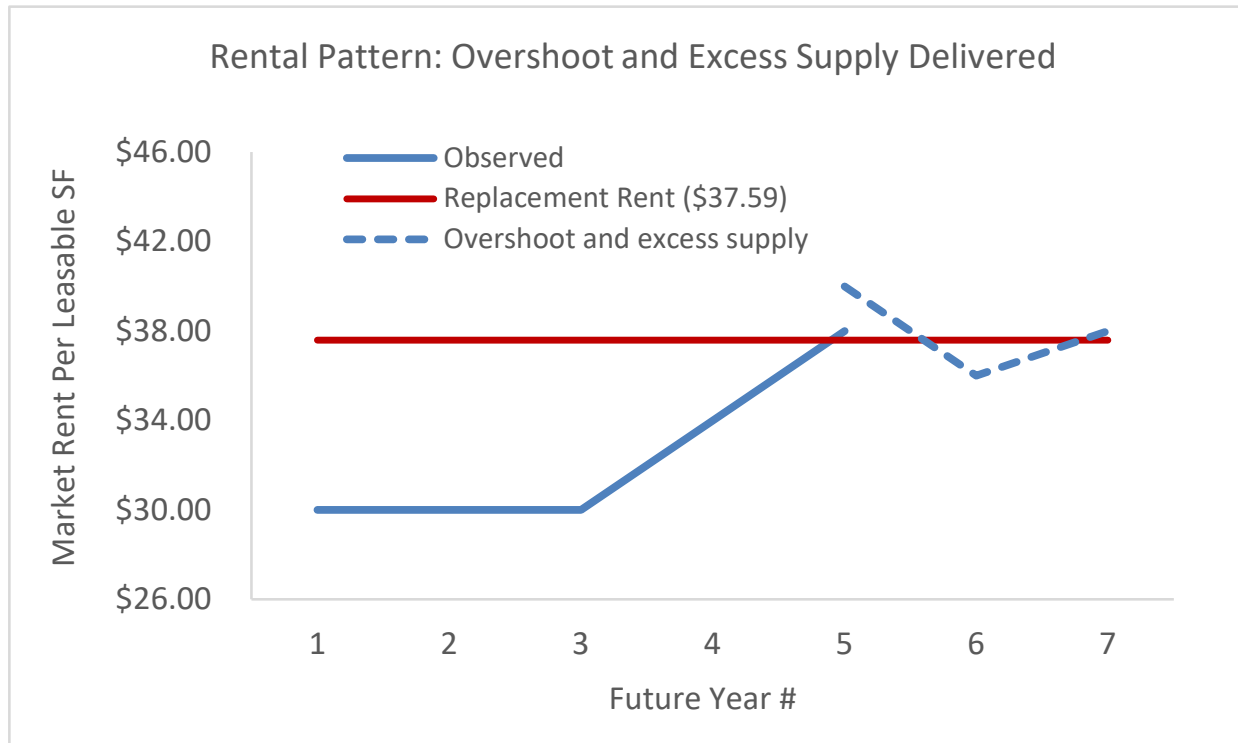


Figure 11.11



Land Costs

- Do not overpay for land
- Solve for your maximum price
- Factor in FAR for finished building

Figure 11.12

Residual Land Valuation of Muk Office Plaza Development Site

Assumptions

Required Return: 10%

Leasable space (Efficiency %): 70%

Vacancy: 5%

Market Rent: \$38.00 per Leasable SF

Effective Rent per Leasable SF: $\$38.00 * (1 - .05) = \36.10

Operating costs per GSF = \$10.00

Soft Costs: \$30.00 per GSF

Hard Costs: \$90.00 per GSF

NOI Calculation

Rent per GSF = Effective Rent per Leasable SF * (1 – Loss Factor %)
 $= \$36.10 * (1 - .30) = \25.27

NOI per GSF = Rental Revenues per GSF – Operating Costs per GSF
 $= \$25.27 - \$10.00 = \$15.27$

Maximum Land Cost Back Solve Calculation

Required Return = NOI / (Soft Costs + Hard Costs + Maximum Land Cost)

Maximum Land Cost = \$x

$10\% = \$15.27 / (\$30.00 + \$90.00 + \$x)$

$10\% * (\$120 + \$x) = \$15.27$

$\$12.00 + (\$x / 10) = \$15.27$

$(\$x / 10) = \3.27

Maximum Land Cost per GSF = \$x = \$32.70

Figure 11.13

Land Price for a 1-Acre Development Site at Differing Floor Area Ratios			
Maximum Land Cost per building GSF	\$32.70	\$32.70	\$32.70
Floor Area Ratio (FAR)	0.5	1	2
Building size	21,780 GSF	43,560 GSF	87,120 GSF
Maximum Price able to pay (Max. Land Cost * GSF)	\$712,206	\$1,424,412	\$2,848,824

Interim Income

- Billboard leasing / signage
- Parking
- Other uses

Figure 11.14

Anoop Court Development Sources and Uses of Funds			
SOURCES OF FUNDS			
	Total	% Total	Per Unit
Equity	7,500,000	25.00%	29,070
Construction Mortgage	<u>22,500,000</u>	<u>75.00%</u>	<u>87,209</u>
Total Sources of Funds	\$30,000,000	100.00%	\$116,279
USES OF FUNDS			
	Total	% Total	Per Unit
Land Acquisition	\$2,326,923	7.76%	\$9,019
Hard Costs			
Rezoning Costs	250,000	0.83%	969
Construction	18,000,000	60.00%	69,767
Contingency	912,500	3.04%	3,537
Total Hard Costs	\$19,162,500	63.88%	\$74,273
Soft Costs			
Architecture & Engineering	2,250,000	7.50%	8,721
Permits	437,000	1.46%	1,694
Utility Fees	255,000	0.85%	988
Owner Allowances	150,000	0.50%	581
Construction Loan Fee	225,000	0.75%	872
Lending Inspection	50,000	0.17%	194
Transfer/Recordation	259,457	0.86%	1,006
Real Estate Taxes	200,000	0.67%	775
Consultants	575,000	1.92%	2,229
Insurance	23,420	0.08%	91
Marketing	500,000	1.67%	1,938
Furnishings, Fixtures & Equipment	350,000	1.17%	1,357
Borrower Legal	50,000	0.17%	194
Lender Legal	250,000	0.83%	969
Development Overhead	1,000,000	3.33%	3,876
Contingency	657,488	2.19%	2,548
Total Soft Costs	\$7,232,365	24.11%	\$28,032
Construction Interest	<u>\$1,278,212</u>	<u>4.26%</u>	<u>\$4,954</u>
Total Uses of Funds	\$30,000,000	100.00%	\$116,279

Figure 11.15

Anoop Court Development Hard Costs Budget Summary			
	Budget	% Total	Per Unit
Hard Costs			
Rezoning Costs	250,000	0.83%	969
Construction	18,000,000	60.00%	69,767
Contingency	912,500	3.04%	3,537
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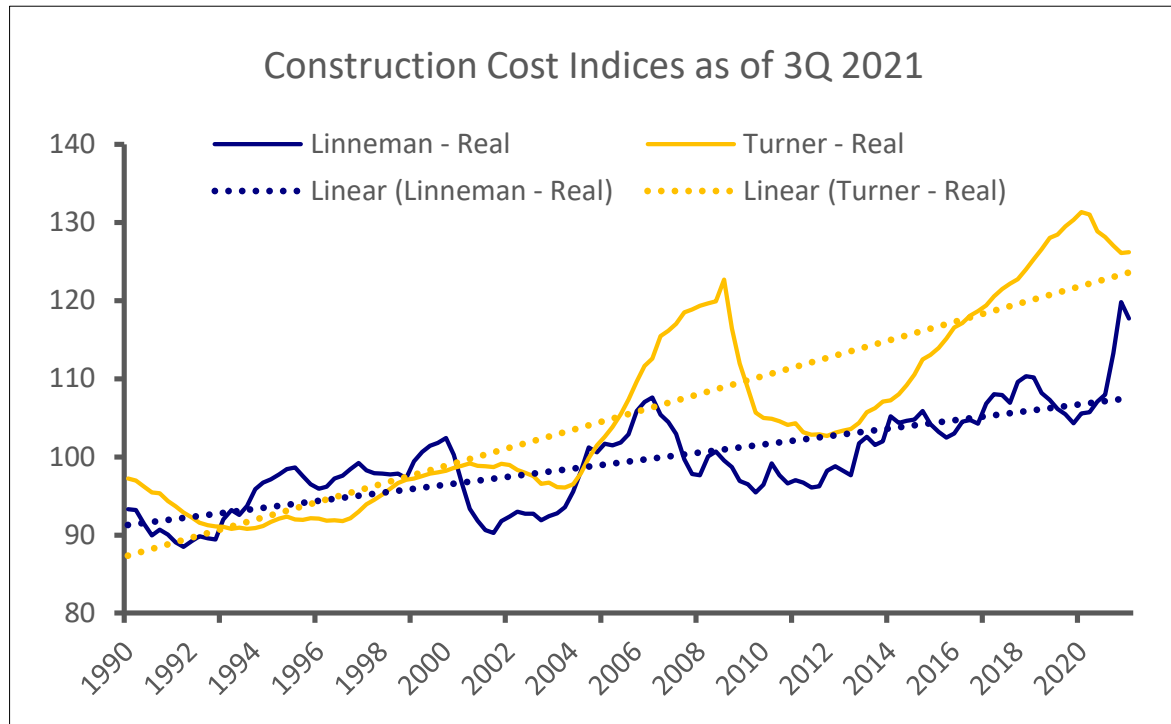
Hard Costs

- Direct Development Costs (related to material & labor):
 - Rezoning costs
 - Construction
 - Architectural Fees

Forecasting Hard Costs

- Market-specific
- Project- and property type-specific

Figure 11.16



Soft Costs

- Indirect development costs
 - Development fees
 - Legal
 - Consultants
 - Transfer & recording fees
 - Contingency reserve
 - Many more

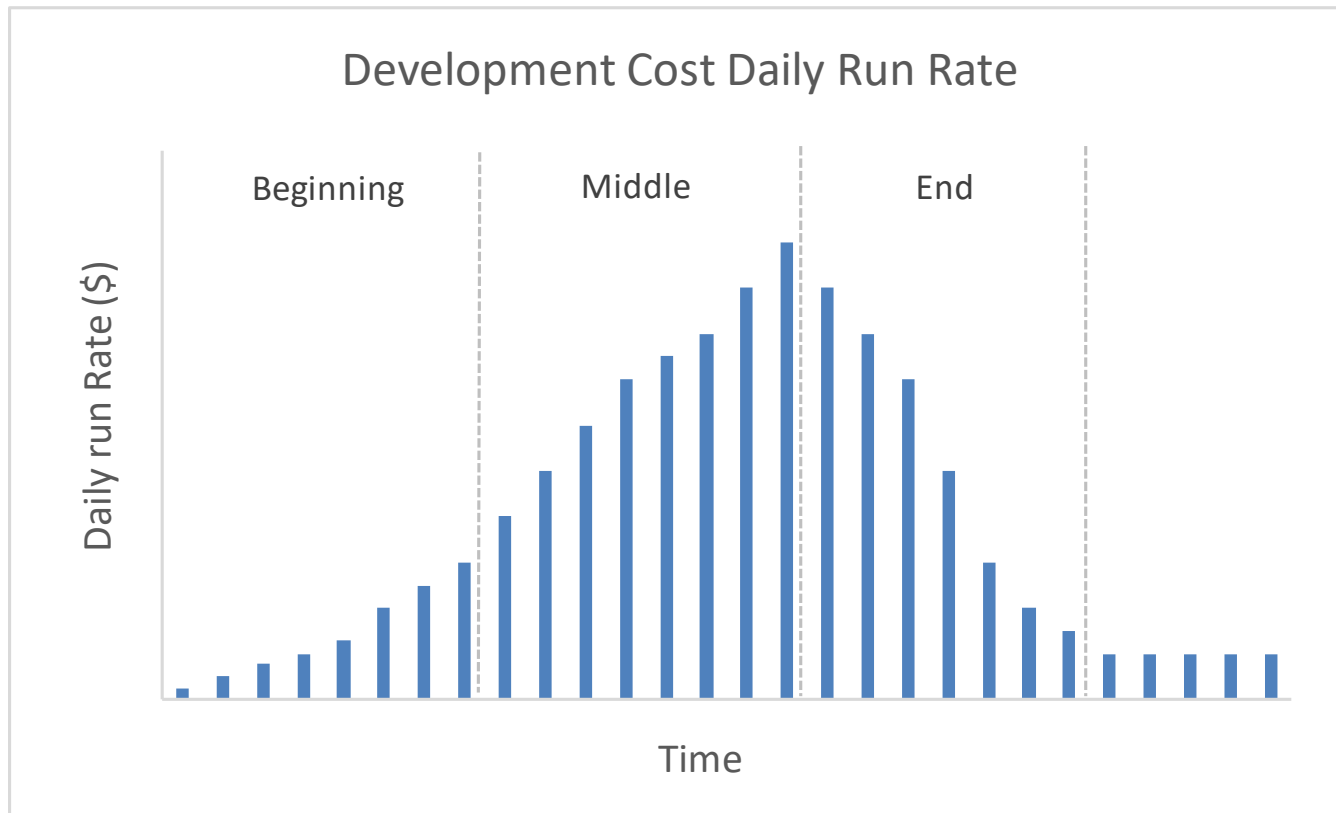
Figure 11.17

Anoop Court Development Soft Costs Budget Summary			
	Budget	% Total	Per Unit
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Contingency	657,488	2.19%	2,548
Total Soft Costs	\$7,232,365	24.11%	\$28,032

Timing

- Capital sources need to know
- Run rate

Figure 11.18



Design

- Impact on costs and revenues
- Impact on loss factor
- Relative to competitive sites